

DEMO: setup a python environment

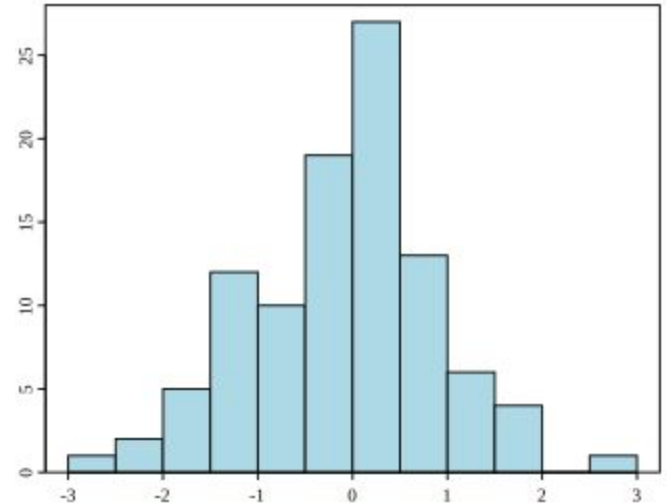
Setup a python environment for Data Science

- **pyenv** to select a python version
- **poetry/pipenv** to create and manage a python environment
- install data viz and data processing libraries:
 - **pandas**
 - **seaborn**

Data Exploration: Histogram

Use case: explore the distribution of a variable

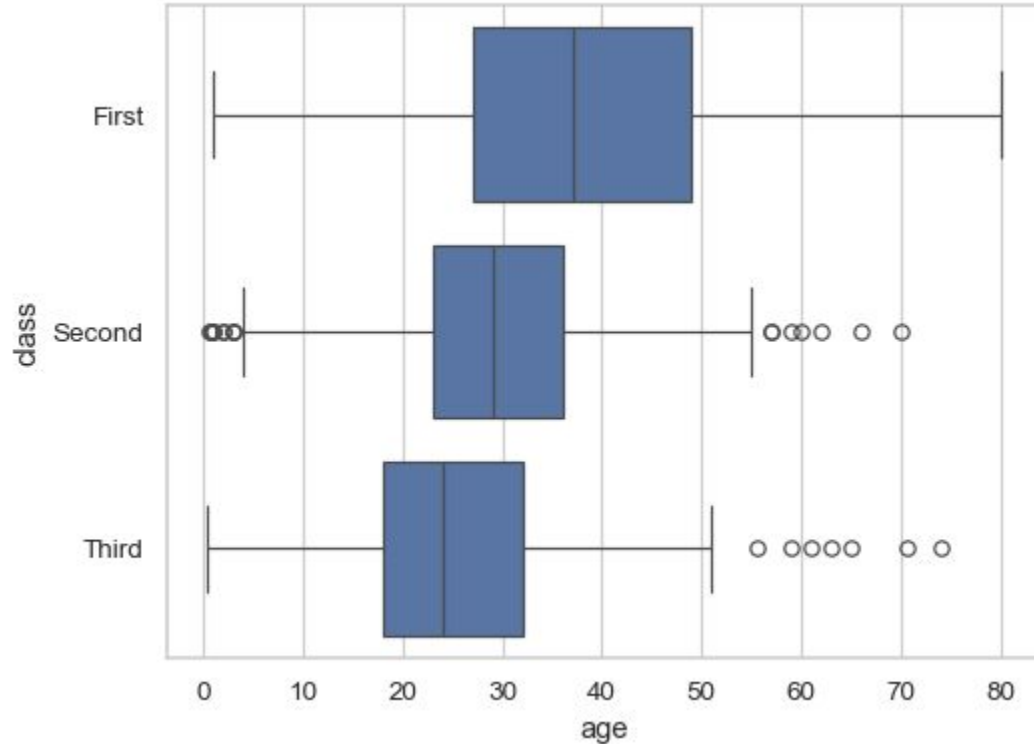
Exercise: distribution of age



Data Exploration: Box Plot

Use case: find the median and extreme values for a variable given another

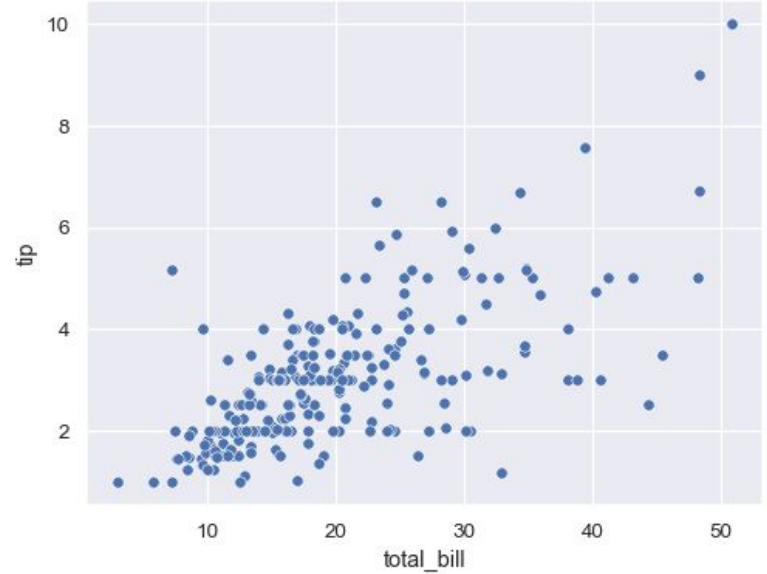
Exercise: distribution of age given class



Data Exploration: Scatter Plot

Use case: see the correlation between two variable

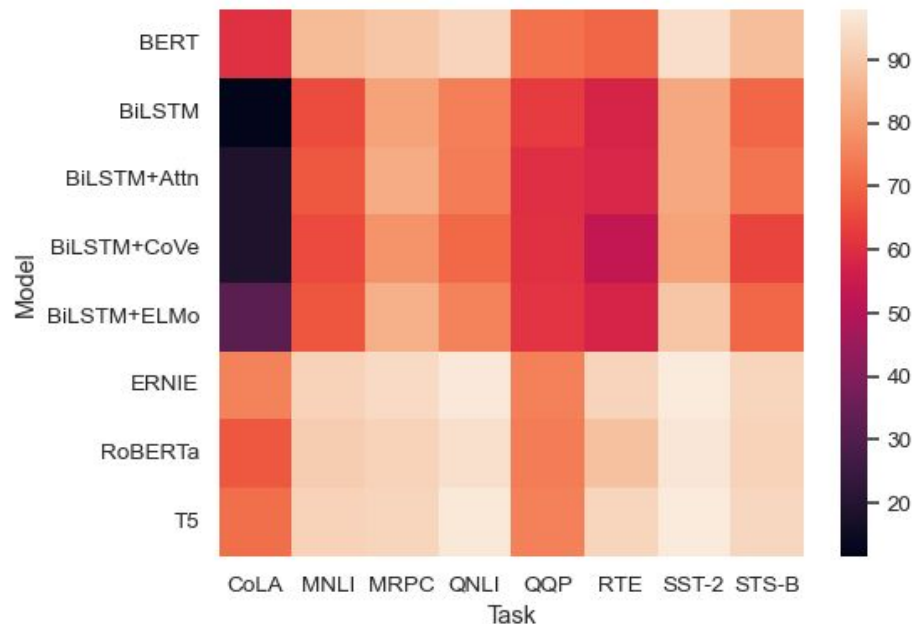
Exercise: age given ticket fare



Data Exploration: Heatmap

Use case: see the correlation between variables

Exercise: heatmap of
Survived, Sex, Fare, Age, Pclass

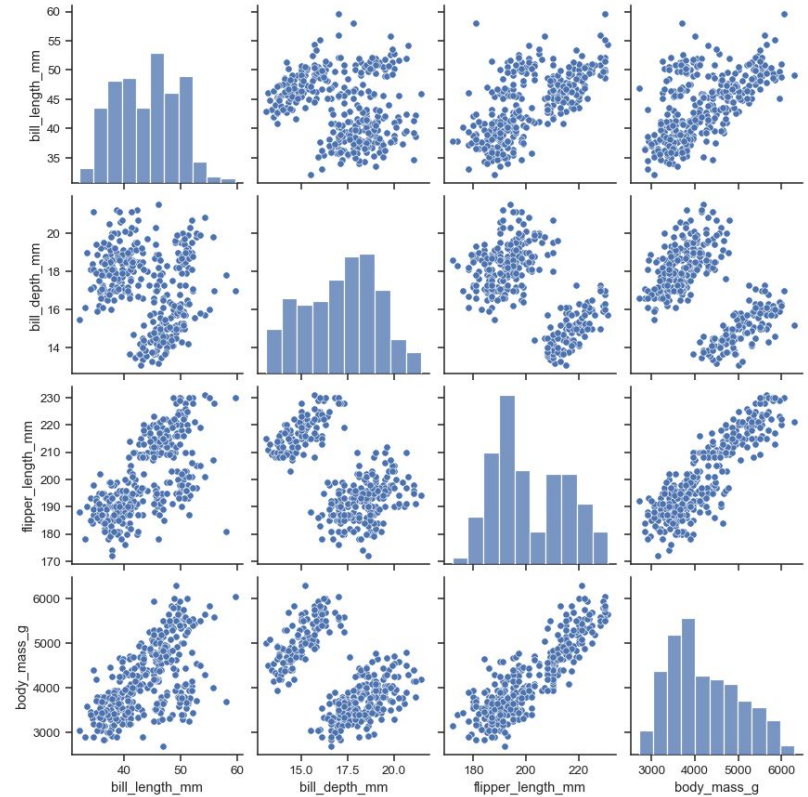


Data Exploration: Pair Plot

Use case: find correlation between variables

Exercise: pairplot of

**Sex, Fare, Age, Pclass given
Survived**



Exercise 1: SpaceShip Titanic Dataset

- 0. setup a python environment
- 1. create an account on [Kaggle](https://www.kaggle.com)
(<https://www.kaggle.com/competitions/spaceship-titanic/overview>)
- 2. download and use jupyter notebook (bonus for correctly configuring your python env using pyenv and poetry)
- 3. download the dataset from kaggle
- 4. produce a histogram plot of passengers age distribution
- 5. produce a boxplot of age given VIP
- 6. produce a scatterplot of age given total sum billed
- 7. heatmap of Transported, Cryosleep, total sum billed, Age, VIP
- 8. pairplot of Transported, Cryosleep, total sum billed, Age, VIP
- 9. create a pie chart of Transported

Session 2

Question Time

- **what's the levels of the wisdom pyramid ? What's the point of the pyramid**

Question Time

- **what's the difference between supervised and unsupervised machine learning ?**

Question Time

- **give an example of training/prediction biais.**

Data Preprocessing

Pre-processing

- the **most time consuming** part of modelisation
- In real data science project, modelisation **represents 20%**
- **absolutely critical**

Agenda

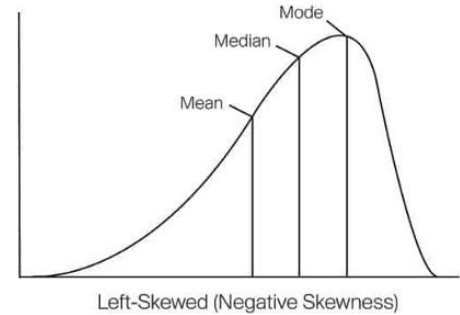
- Handling Issues With Datasets
- Scaling Data
- Encoding Data
- Splitting A Dataset For Training And Testing

Handling Missing Data

- remove the row
- remove the column

Handling Missing Data

- replace value with mean, median or mode

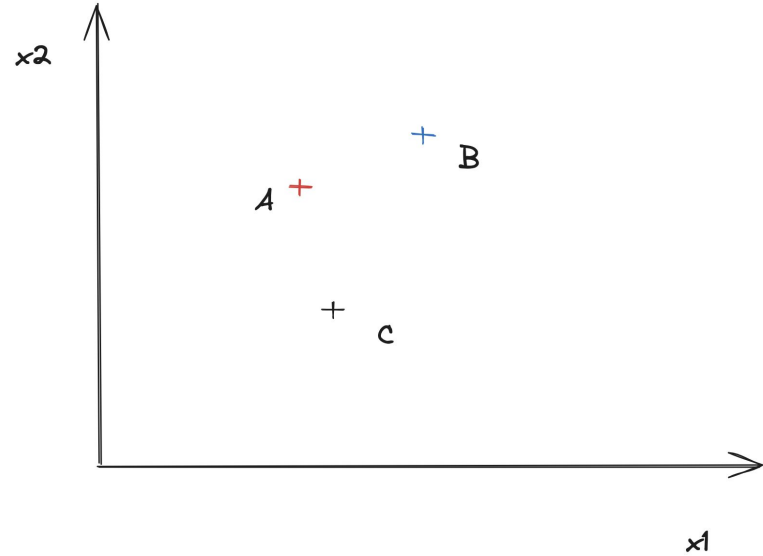


Handling Missing Data

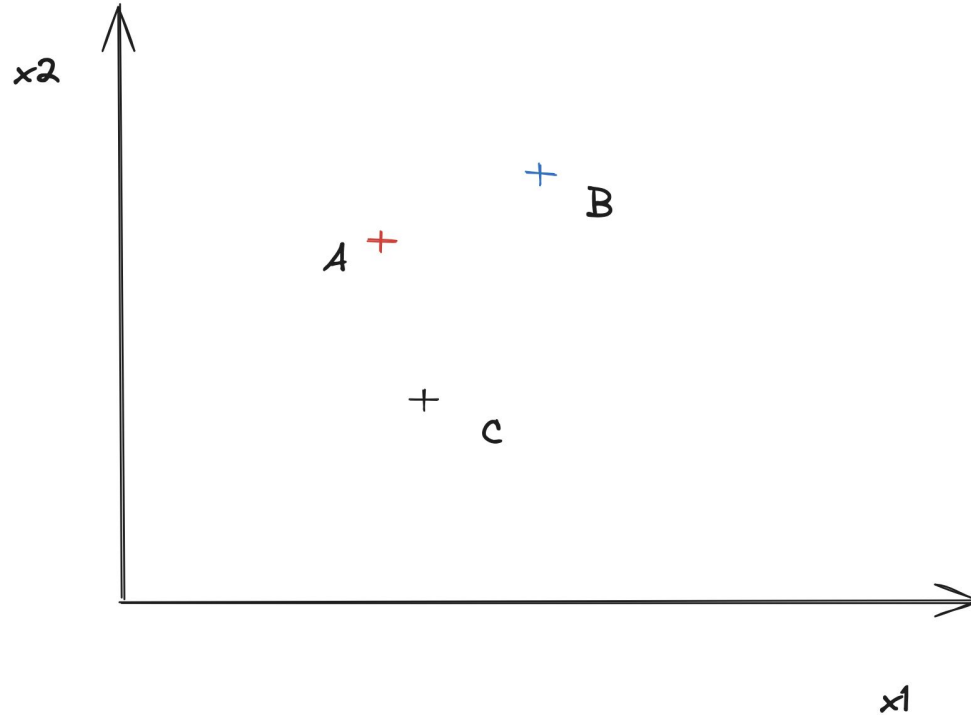
- if categorical, create a “MISSING” category

Handling Missing Data

- use KNN imputer



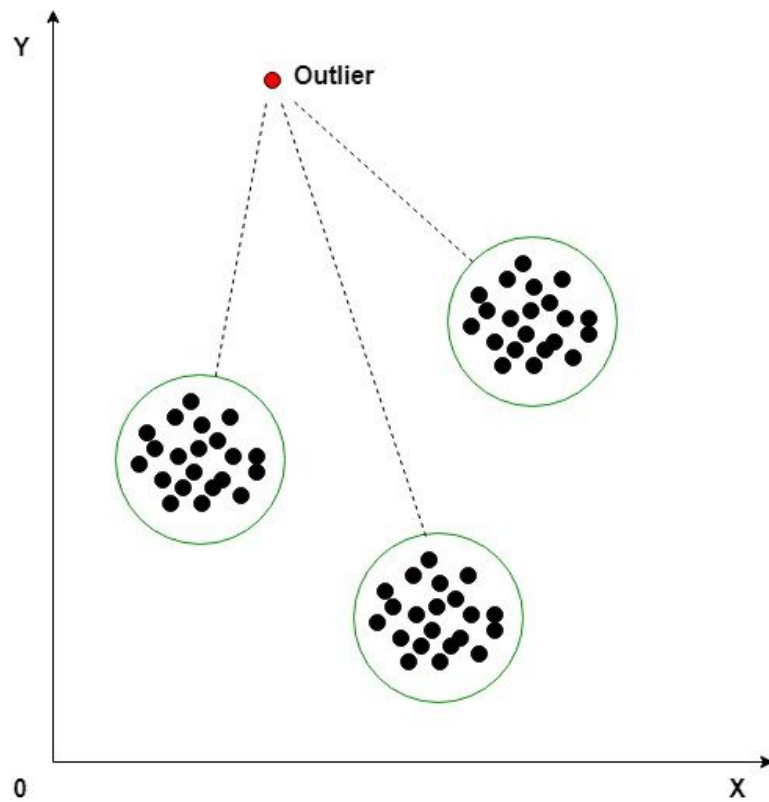
KNN - Example - Exercice



Handling Missing Data

- use a model that manages missing values

outlier



Dealing with Outliers

Some models, normalisation or scaling methods are **sensible to outliers**. To Detect them:

- using Data Viz to technique
- use **anomaly detection**

To handle them:

- **use robust models**
- **remove them ?**
- **apply specific data transformation to handle them**

Feature Scaling

- Some models are sensitive to the scale of a feature (logistic regression, KNN)

- **Min-Max Scaling**

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Using Min-Max Scaler with Titanic Dataset

DEMO

Min Max Scaling

- **sensitive to outliers**
- **works for every kind of distribution**
- **simple**